Microeconomics I

Class 8 — Choice under uncertainty: expected utility and risk aversion

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Plan for today

Warm-up; two choices, by hands up

Lotteries and the axioms

Expected utility: form and theorem

In pairs: reduce and rank

Break

Your two choices, revisited; uniqueness

Money lotteries and risk aversion

Certainty equivalent and risk premium

In pairs: one game, three people

Summary

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Class 8 opens the second block of the course. Until now the objects of choice were bundles of goods, received for sure; from today they are lotteries, lists of probabilities over prizes. The first hour asks which preferences over lotteries can be written as an expected utility; the second hour puts money on the prizes and asks what it means to dislike risk, and how to measure that dislike in euros. MWG 6.B–6.C.

Slide 2 Plan for today

Before the break: a warm-up on class 7, a pair of choices decided by a show of hands, and then the building blocks: lotteries as probability vectors, compound lotteries, the two new axioms (continuity and independence), the expected utility form and the theorem that delivers it, with an exercise on reducing and ranking. After the break: the two choices looked at again with the new tools; why an expected utility function is unique only up to a positive affine transformation; then money lotteries, risk aversion, the certainty equivalent and the risk premium, with a second exercise.

MWG 6.B–6.C; Muñoz-Garcia chapter 5 is the gentler read.

Practicalities

- Problem set 2: handed in at the start of class, Wed 21 Oct; returned graded on Tue 27 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Slide 3 Practicalities

Problem set 2 was handed in at the start of class on Wednesday 21 October, on paper, one copy per group; it comes back graded on Tuesday 27 October. Nothing from today's class is on it.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: always true, or not?

True for every consumer?

p_1 rises: own substitution effect ≤ 0	Inferior good $\Rightarrow$ Giffen good
compensating variation (CV) $=$ equivalent variation (EV)	$\frac{\partial h_1}{\partial p_2} = \frac{\partial h_2}{\partial p_1}$

Class 7, slides 9, 12, 16, 20. Mas-Colell, Whinston and Green (MWG) Prop. 3.G.2, 3.G.3; (3.I.1)–(3.I.2).

Slide 4 Warm-up: always true, or not?

Four one-line claims from class 7, one for each of its main results: the sign of an own substitution effect, the relation between Giffen and inferior goods, the two money measures of a price change, and the cross effects of compensated demand. Before class, decide for each whether it holds for every consumer and name the result it rests on.

Answers revealed in class. For each claim, ask whether it is about the substitution part of a price change or about its wealth part: the first obeys general laws, the second depends on the good.

Two choices: which would you take?

Imagine the prizes are real and paid once. No right answer.

First choice

A: 1 million euros for sure

B: 10 %: 5 million euros
89 %: 1 million euros
1 %: nothing

Second choice

C: 11 %: 1 million euros
89 %: nothing

D: 10 %: 5 million euros
90 %: nothing

Hands up: A or B? Then: C or D?

A classic pair of choices; the source in class.

Slide 5 Two choices: which would you take?

Two pairs of hypothetical prizes, decided by a show of hands before any theory. There is no right answer: each choice on its own is a matter of taste. The counts come back after the break, when the room's own votes are the data.

Before class, decide honestly what you would take in each pair; then compare A with B, and C with D, prize by prize, and ask what the two lotteries in each pair have in common. The source comes in class.

Lotteries and expected utility

MWG 6.B

Slide 6 Lotteries and expected utility

Part 1: lotteries, the axioms, and the expected utility form. MWG 6.B.

A simple lottery is a list of probabilities: a point in a triangle

Simple lottery: $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$; p_n = probability of outcome n.

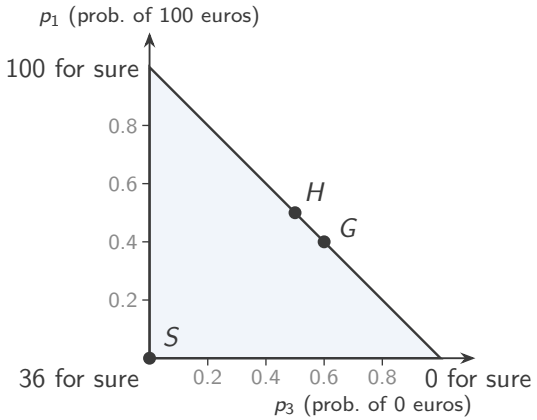
Outcomes: 100, 36, 0 euros.

$S = (0, 1, 0)$: 36 euros for sure

$H = (0.5, 0, 0.5)$: a coin, 100 or 0

$G = (0.4, 0, 0.6)$: 40 % chance of 100

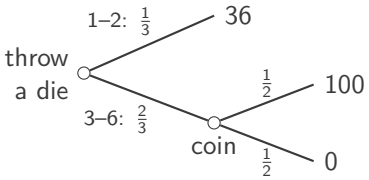
MWG Def. 6.B.1, p. 168; Fig. 6.B.1, p. 169 draws the same set as an equilateral triangle.



$p_2 = 1 - p_1 - p_3$: 0 on the hypotenuse, 1 at S

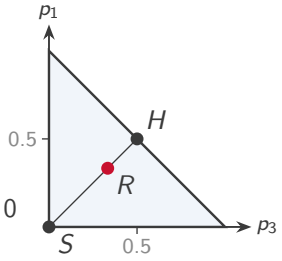
A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Compound lottery $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$: you get L_k with probability α_k.
Reduced lottery: $\alpha_1 L_1 + \cdots + \alpha_K L_K$.



$R = (S, H;\ \frac{1}{3}, \frac{2}{3})$

$\frac{1}{3}(0, 1, 0) + \frac{2}{3}(0.5, 0, 0.5)$
$= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
= one throw: 1–2 → 100,
3–4 → 36, 5–6 → 0



Only the reduced lottery matters (MWG's consequentialist premise): the die-then-coin tree and one throw are the same lottery.

MWG Def. 6.B.2, pp. 169–170; Figs. 6.B.2–6.B.3, p. 170.

Slide 7 A simple lottery is a list of probabilities: a point in a triangle

A lottery is described by its probabilities, not by its prizes: the prizes are fixed once and for all (here 100, 36 and 0 euros, in this order), and a lottery is the vector that says how likely each one is. The entries of the vector are probabilities, so they lie between 0 and 1 and add up to 1; a vector of prizes is not a lottery, which is the first notation trap of the block.

With three prizes the third probability is fixed by the other two, so every lottery is a point of a triangle: the corners are the sure prizes, each side holds the lotteries that never pay one of the prizes, and the hypotenuse here holds those that never pay 36. The three named lotteries S, H and G are used all morning. MWG draws the same set as an equilateral triangle (Fig. 6.B.1, p. 169); flattening it changes nothing.

Slide 8 A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

A compound lottery is a lottery whose prizes are themselves lotteries. The rule for reducing it follows from how probabilities combine along a path: multiply along the branches, then add over the paths that end in the same prize. Here the coin is reached only two times in three, which is why the 100 and the 0 each get $\frac{2}{3} \cdot \frac{1}{2}$ and not $\frac{1}{2}$; adding probabilities across the stages is the slip to avoid.

In vectors the reduced lottery is the weighted average of the component lotteries, $\frac{1}{3}S + \frac{2}{3}H$, which in the triangle is the point two thirds of the way from S to H. Every result that follows uses this linear structure. The premise of the whole theory (MWG p. 170) is that only the reduced lottery matters: two trees with the same reduction are the same object of choice.

Rational and continuous preferences over lotteries

Rational: complete, transitive (class 1).

Continuous: for all $L, L', L'' \in \mathcal{L}$ the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha \in [0,1] : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed.

Small probability changes never reverse a strict ranking.

> **Fails: safety first** (no chance of 0 beats any chance).
> $L = 100$, $L' = 0$, $L'' = 36$ euros for sure: second set $= [0, 1)$.

MWG Def. 6.B.3, p. 171 ("safety first": same page). Class 3: lexicographic order, not continuous.

Independence: mixing both with the same L'' keeps their ranking

Independence axiom: for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0,1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$



$\alpha L + (1-\alpha)L''$ $\qquad$ $\alpha L' + (1-\alpha)L''$

Tails gives both the same L'', and you get it instead of L or L', never with it: unlike bundles of goods.

MWG Def. 6.B.4, p. 171 (fn. 5: von Neumann and Morgenstern, 1944); coin-toss reading and Fig. 6.B.4, p. 172.

Slide 9 Rational and continuous preferences over lotteries

Rationality is the same pair of axioms as in class 1, now on the set of lotteries; MWG notes that it asks more here, because comparing risky alternatives is harder than comparing bundles (p. 171). Continuity is written with mixtures: the sets of weights α for which the mix is at least as good as a third lottery, or at most as good, must be closed. In words, a small change in the probabilities never turns a strict preference around; MWG's example is a car trip with a tiny chance of an accident, still better than staying at home.

The counterexample is class 3's lexicographic idea in lottery form: a consumer who puts safety first ranks any lottery with no chance of 0 above any lottery with one. The card names the instance that fails: with L and L' the sure 100 and the sure 0, and L'' the sure 36, the set of weights for which 36 is at least as good is $[0, 1)$, which misses its limit point. As in class 3, continuity is what makes a utility function on lotteries exist at all; independence, next, is what gives it a formula. The two axioms are easy to mix up: continuity is about limits, independence about mixing with a common third lottery.

Slide 10 Independence: mixing both with the same L'' keeps their ranking

The axiom that makes the theory: a ranking of two lotteries survives when both are mixed, with the same weight, with the same third lottery. The coin picture explains why one would accept it: on tails both coins pay the same L'', so nothing differs; on heads one coin pays L and the other L'; so the coin with the preferred lottery should be preferred, whatever L'' is.

Why this is a genuinely new assumption, and not class 1 again, is in MWG's remark on p. 172: with goods, how much you like apples can depend on the pears you also get, because you consume them together; here L'' comes instead of L, never with it. Read the quantifiers: the axiom asks for every α strictly between 0 and 1 and every L''. It also holds with strict preference and with indifference in place of the weak ranking (MWG Exercise 6.B.1, p. 172), which Exercise 1 uses.

Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

U has the expected utility form if there are numbers $u_1, \ldots, u_N$ with $U(L) = u_1 p_1 + \cdots + u_N p_N$ for every L.

Such a U is a von Neumann–Morgenstern (vNM) expected utility function.

Her utilities for 100, 36, 0 euros: 10, 6, 0.

$U(S) = 6, \quad U(H) = 5$

$U(R) = \frac{1}{3} \cdot 10 + \frac{1}{3} \cdot 6 + \frac{1}{3} \cdot 0 = \frac{16}{3} = \frac{1}{3}\, U(S) + \frac{2}{3}\, U(H)$

MWG Def. 6.B.5 and Prop. 6.B.1 (expected utility form $\iff$ U linear), p. 173.

Slide 11 Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

The expected utility form attaches one number u_n to each prize and values a lottery by the probability-weighted average of those numbers: the same operation as the expected value $E[x] = \sum_n p_n x_n$, with utilities in place of euros. The notation trap is the pairing: each probability multiplies the utility of its own prize, as in the dot product $p \cdot u$.

Her utilities 10, 6, 0 for 100, 36 and 0 euros give the first sign of risk aversion: she prefers 36 for sure to a fair coin for 100 or 0, although the coin's expected prize is 50 euros. The die lottery R of slide 8 shows the property that matters: its utility is the weighted average of the utilities of its parts. That is MWG Prop. 6.B.1, p. 173: having the expected utility form is the same as being linear in the probabilities.

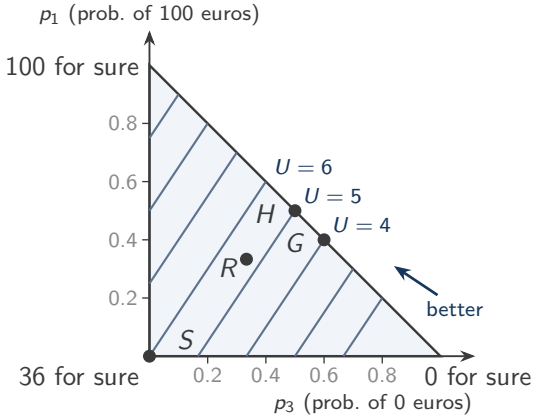
Continuity + independence ⇒ expected utility: straight, parallel lines

Expected utility theorem: if $\succsim$ on $\mathcal{L}$ is rational, continuous and satisfies independence, there are $u_1, \ldots, u_N$ such that for all L, L':
$L \succsim L' \iff \sum_n u_n p_n \geq \sum_n u_n p'_n$.

In the triangle: indifference curves are straight, parallel lines.

Conversely, every expected utility U satisfies independence (MWG Ex. 6.B.2).

MWG Prop. 6.B.3, (6.B.4), p. 176; proof pp. 176–178; Fig. 6.B.5, p. 175.



Slide 12 Continuity + independence ⇒ expected utility: straight, parallel lines

The theorem says the three axioms are exactly what an expected utility representation needs: rational, continuous and independent preferences over lotteries can be written as $\sum_n u_n p_n$, and (this slide's last line, and Optional A) any such U satisfies the axioms. The proof idea is short. By continuity every lottery is exactly as good as some mix of the best prize and the worst, and the weight on the best prize can serve as its utility; independence then forces this utility to be linear in the probabilities. The details are in MWG pp. 176–178.

The picture is the test to remember: a linear U has level sets that are straight lines with one common slope, so the indifference map of an expected utility maximizer in the triangle is a family of straight, parallel lines, here of slope 1.5. Curved or fanning indifference lines are not a different taste; they contradict independence (MWG Fig. 6.B.5, p. 175). The trap is to draw them bowed, like indifference curves over goods.

In pairs: reduce, then rank without computing

Outcomes 100, 36, 0 euros; her utilities 10, 6, 0.
$S = (0, 1, 0)$, $G = (0.4, 0, 0.6)$, $H = (0.5, 0, 0.5)$.

$K = (G, H;\ \frac{1}{2}, \frac{1}{2})$, $\quad K' = (S, H;\ \frac{1}{2}, \frac{1}{2})$

Per pair: three answers, three minutes.

(1) Reduce K.

(2) $U(K)$.

(3) K or K'? Which axiom?

Answers revealed in class. Try it in pairs before class.

MWG Defs. 6.B.2, 6.B.4; Exercise 6.B.1, p. 172 (strict version).

Slide 13 In pairs: reduce, then rank without computing

Two compound lotteries that share a coin and one of its branches; her utilities for the three prizes are on the slide. Part (1) is slide 8's rule for reducing a compound lottery; part (2) can be done two ways, directly or by linearity, and they must agree; part (3) asks for a ranking with no computation at all, from one of the two new axioms.

Try it in pairs before class, and for (3) say which branches of the two lotteries differ and which do not. Answers revealed in class.

(optional) Expected utility form ⇒ independence

Show: if U has the expected utility form, then $\succsim$ satisfies the independence axiom.

Hint: by Prop. 6.B.1, $U(\alpha L + (1-\alpha)L'') = \alpha\, U(L) + (1-\alpha)\, U(L'')$.

At home if not done in class; the answer is in the slides released after class.

MWG Exercise 6.B.2, p. 175.

Slide 14 (optional) Expected utility form ⇒ independence

An optional exercise, to be done at home if class time runs out: show that a preference relation with an expected utility representation satisfies the independence axiom. The hint on the slide is the linearity of U from slide 11; write both sides of the axiom in terms of U and compare them.

The answer is in the slides and handout released after class. Watch the role of α: ask what happens to the comparison when $\alpha = 0$.

Break

Ten minutes

slide 15 of 25

Slide 15 Break

Ten minutes.

The two choices, revisited

Revealed in class

Before class: in each pair, what do the two lotteries have in common?

slide 16 of 25

Slide 16 The two choices, revisited

Revealed in class.

Before class: in each pair of slide 5, what do the two lotteries have in common? Write each of the four as a list of probabilities over the three prizes before comparing.

Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; u^2 flips her choice

utilities of (100, 36, 0 euros)	$U(G)$ $G = 40\,\%$ of 100	$U(S)$ $S = 36$ for sure	she picks
$u = (10, 6, 0)$	4	6	S
$2u + 3 = (23, 15, 3)$	11	15	S
$u^2 = (100, 36, 0)$	40	36	G

$\tilde{U}$ is another vNM utility for $\succsim \iff \tilde{U} = \beta U + \gamma$ with $\beta > 0$.

u^2 keeps the order of prizes, not lotteries:
unlike class 3, expected utility is cardinal.

MWG Prop. 6.B.2, pp. 173–174.

Slide 17 Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; u^2 flips her choice

Class 3 said that utility over bundles is ordinal: any increasing transformation represents the same preferences. Over lotteries, an increasing transformation of U still ranks lotteries alike but is no longer an expected utility; an increasing transformation of the prize utilities u_n can reverse a choice. Doubling and adding 3 rescales every lottery's value alike, so the choice between G and S stays; squaring the prize utilities keeps the sure prizes in order yet reverses the choice, because an average of squares is not the square of an average. So an expected utility function is unique up to $\beta U + \gamma$ with $\beta > 0$ (MWG Prop. 6.B.2, pp. 173–174): it is cardinal.

The squared person's utilities are 100, 36 and 0, the prizes themselves: this consumer ranks lotteries by expected money and returns on slide 19. The trap is the class 3 habit of applying any increasing f to the u_n: only $\beta U + \gamma$ keeps both the ranking and the expected utility form.

Money lotteries and risk aversion

MWG 6.C

Slide 18 Money lotteries and risk aversion

Part 2: money lotteries, risk aversion, the certainty equivalent and the risk premium. MWG 6.C.

A money lottery is a distribution function F; $U(F) = \int u(x)\,dF(x)$

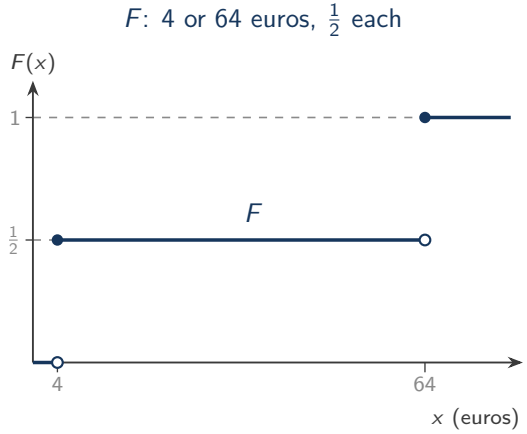
$F(x)$ = probability that the payoff is at most x.

$U(F) = \int u(x)\,dF(x)$; with prizes x_n and probabilities p_n: $\sum_n p_n\,u(x_n)$.

U: vNM utility, on lotteries.
u: Bernoulli utility, on money — increasing, continuous.

Her 10, 6, 0 for 100, 36, 0 euros: $u(x) = \sqrt{x}$.

MWG p. 183; (6.C.1) and fn. 12, p. 184; p. 185.



Slide 19 A money lottery is a distribution function F; $U(F) = \int u(x)\,dF(x)$

With money prizes a lottery can have any number of outcomes, so it is described by its distribution function, the probability of receiving at most x; for the lottery on the slide F is 0 below 4 euros, jumps by a half at 4 and again at 64. Expected utility is the same formula as on slide 11, written as an integral; with finitely many prizes it is the sum $\sum_n p_n u(x_n)$, the expected value of $u(x)$.

Two functions carry the name utility, and the course keeps them apart as MWG does (fn. 12, p. 184): U, the von Neumann–Morgenstern utility, takes a lottery; u, the Bernoulli utility, takes an amount of money, and it is increasing and continuous. Calling u the vNM utility is the naming slip to avoid. Her numbers from the first hour were $\sqrt{x}$ at 100, 36 and 0 all along, and the squared person of slide 17 has $u(x) = x$; both return on the next slides.

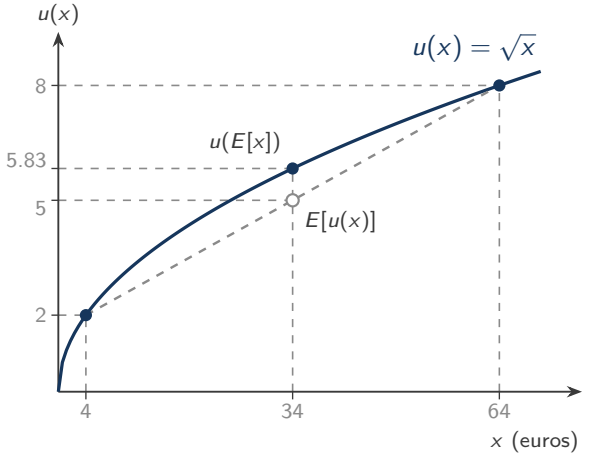
Risk averse ⇔ u concave: the chord lies below the graph

Risk averse: for every F, the sure amount $\int x\,dF(x)$ is at least as good as F.

With expected utility: $\int u(x)\,dF(x) \le u(\int x\,dF(x))$ for every F — Jensen: u concave.

Concave in money: a property of the function u, not convexity of preferences over bundles (classes 2–3). U is linear in probabilities.

MWG Def. 6.C.1 and (6.C.2), p. 185; Fig. 6.C.2, p. 186.



Slide 20 Risk averse ⇔ u concave: the chord lies below the graph

Risk aversion is defined without any utility function: for every lottery, its expected value paid for sure is at least as good as the lottery itself (MWG Def. 6.C.1, p. 185); risk neutral means always indifferent, strictly risk averse means the sure amount is strictly better whenever the lottery is not degenerate. For an expected utility maximizer the definition becomes Jensen's inequality, the average of u at most u at the average, for every F, and that is exactly concavity of u: the chord lies below the graph. The figure reads it on slide 19's lottery: the chord's midpoint is $E[u(x)] = 5$, the curve at $E[x] = 34$ is about 5.83.

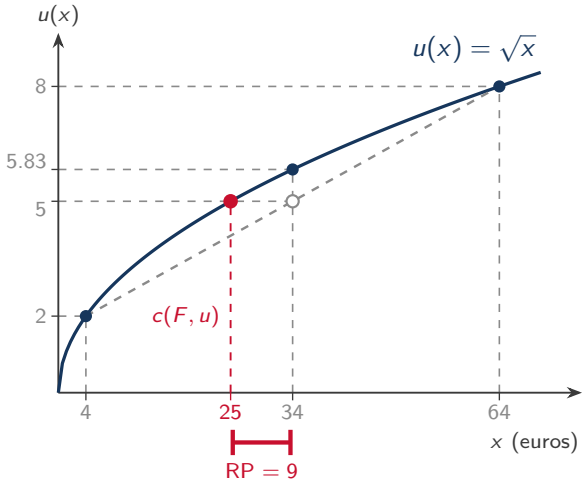
Two re-anchorings from classes 2 and 3 matter here. This is concavity of a function of money, checked by the chord test; it is not convexity of preferences over bundles and not quasiconcavity, and U itself is always linear in the probabilities, so risk aversion is never about curvature in p. The boundary trap of the block is to write $u(E[x])$ for $E[u(x)]$: the first is a point on the curve, the second a point on the chord, and they are equal only when u is linear.

Certainty equivalent 25 euros, risk premium 9 euros: three steps

Certainty equivalent $c(F, u)$:
$u\big(c(F,u)\big) = \int u(x)\, dF(x)$.
Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

MWG Def. 6.C.2(i), (6.C.3), p. 186: "some expected return is traded for certainty". RP as in Muñoz-Garcia, p. 357.



In pairs: one game, three people

Toss two coins. Both heads: 64 euros. Otherwise: nothing.

In pairs, one table per pair: four numbers per row. Four minutes.

	Bernoulli u	$E[x]$ euros	$E[u(x)]$ utils	$c(F,u)$ euros	RP euros
Ana	$\sqrt{x}$				
Ben	x				
Carla	x^2				

Answers revealed in class. Try it in pairs before class.

The three steps of slide 21.

Slide 21 Certainty equivalent 25 euros, risk premium 9 euros: three steps

The certainty equivalent is the sure amount she finds exactly as good as the lottery, defined by $u(c) = E[u(x)]$ (MWG Def. 6.C.2(i), p. 186), and the risk premium is the expected value she gives up to be rid of the risk, $E[x] - c(F,u)$. The recipe never changes: the expected value, in euros; the expected utility, in utils; then back to euros through u^{-1}. Here 5 utils correspond to 25 euros, so she would swap a lottery worth 34 euros on average for 25 sure euros, a premium of 9 euros.

The three slips all cross a boundary between the steps: using $u(E[x])$, about 5.83, in place of $E[u(x)] = 5$; reporting the certainty equivalent as 5, a number of utils, instead of 25 euros; and flipping the sign of the premium. MWG does not name the risk premium and only says that some expected return is traded for certainty; Muñoz-Garcia (p. 357) names it, as $E[x] - c(F,u)$, and the equivalent form $u(E[x] - \mathrm{RP}) = E[u(x)]$ says the same thing. In the picture the certainty equivalent is where the curve reaches the chord's height, to the left of the mean for a concave u.

Slide 22 In pairs: one game, three people

One game and three people who differ only in their Bernoulli utility: $\sqrt{x}$, x and x^2. For each, fill in the four columns with the three steps of slide 21; the expected value of the game comes first and is the same for all three, since it does not depend on u. Try it in pairs before class.

Before computing, predict the sign of each person's risk premium from the shape of u, then check. Keep the units straight in every cell, euros or utils. Answers revealed in class.

One game, three people: a picture

Revealed in class

Do slide 22 first: the picture shows its answers.

Slide 23 One game, three people: a picture

Revealed in class.

Do slide 22 first: the picture shows its answers. Before class, think about why rescaling each u by a positive affine transformation, which slide 17 allows, is harmless here.

(optional) Ana is richer: what is the game worth to her now?

Ana, $u(x) = \sqrt{x}$, has 36 euros plus slide 22's game (64 euros, probability $\frac{1}{4}$).

(1) $E[x]$, $E[u(x)]$ of her wealth.

(2) $c(F, u)$, and the game's value.

(3) RP.

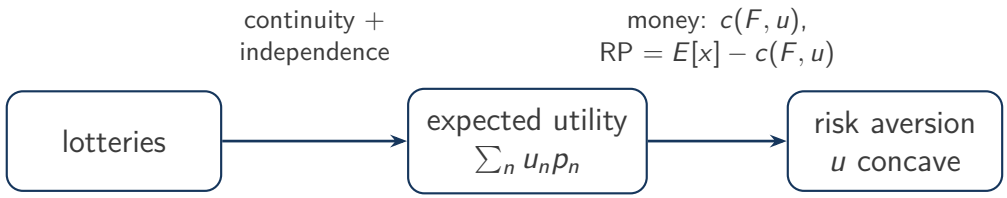
At home if not done in class; the answers are in the slides released after class.

Slide 24 (optional) Ana is richer: what is the game worth to her now?

An optional exercise, to be done at home if class time runs out: Ana from slide 22, now with 36 euros for sure on top of the two-coin game. Her wealth is a lottery with two outcomes; apply the three steps of slide 21 to that lottery, not to the game alone, and then ask what the game adds to a sure 36 euros.

The answers are in the slides and handout released after class. Before computing, say whether you expect the game to be worth more or less to her than on slide 22, and why.

Today in one line



Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance. First- and second-order stochastic dominance, mean-preserving spreads. Review.

Read MWG 6.C–6.D (or Muñoz-Garcia ch. 5) for Tuesday 27 October.

Slide 25 Today in one line

In one line: a lottery is a vector of probabilities, and a compound lottery is the weighted average of its parts; with continuity and independence, preferences over lotteries are an expected utility, straight and parallel in the triangle, and unique up to a positive affine transformation, so that expected utility is cardinal; with money prizes, risk aversion is concavity of the Bernoulli utility, and the certainty equivalent and the risk premium measure it in euros through the three steps.

Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance; first- and second-order stochastic dominance, mean-preserving spreads; review. Reading: MWG 6.C–6.D, or Muñoz-Garcia chapter 5.